203 Project

2026-03-29

```r
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(lmtest)
```

```
## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(sandwich)
library(ggplot2)
library(modelsummary)
```

```
## Warning: package 'modelsummary' was built under R version 4.5.3
```

```r
library(tidyr)

df <- read_excel('../data/203_projectdata.xlsx', sheet = '2022')

# check missing GDP per capita
sum(is.na(df$gdp_pcap))
```

```
## [1] 2
```

```r
# drop missing rows

df_clean <- df %>%
  filter(!is.na(gdp_pcap))

# check missing rows again
sum(is.na(df_clean$gdp_pcap))
```

```
## [1] 0
```

```r
sum(is.na(df_clean$lex))
```

```
## [1] 0
```

Base model Life expectancy vs. GDP per capita

```r
model_base <- lm(lex ~ gdp_pcap, data = df_clean)
summary(model_base)
```

```
##
## Call:
## lm(formula = lex ~ gdp_pcap, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.3046  -3.7819   0.8614   4.9699  11.2529
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 6.966e+01  5.409e-01  128.78  < 2e-16 ***
## gdp_pcap    9.539e-05  1.142e-05    8.35 1.37e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.238 on 191 degrees of freedom
## Multiple R-squared:  0.2674, Adjusted R-squared:  0.2636
## F-statistic: 69.72 on 1 and 191 DF,  p-value: 1.367e-14
```

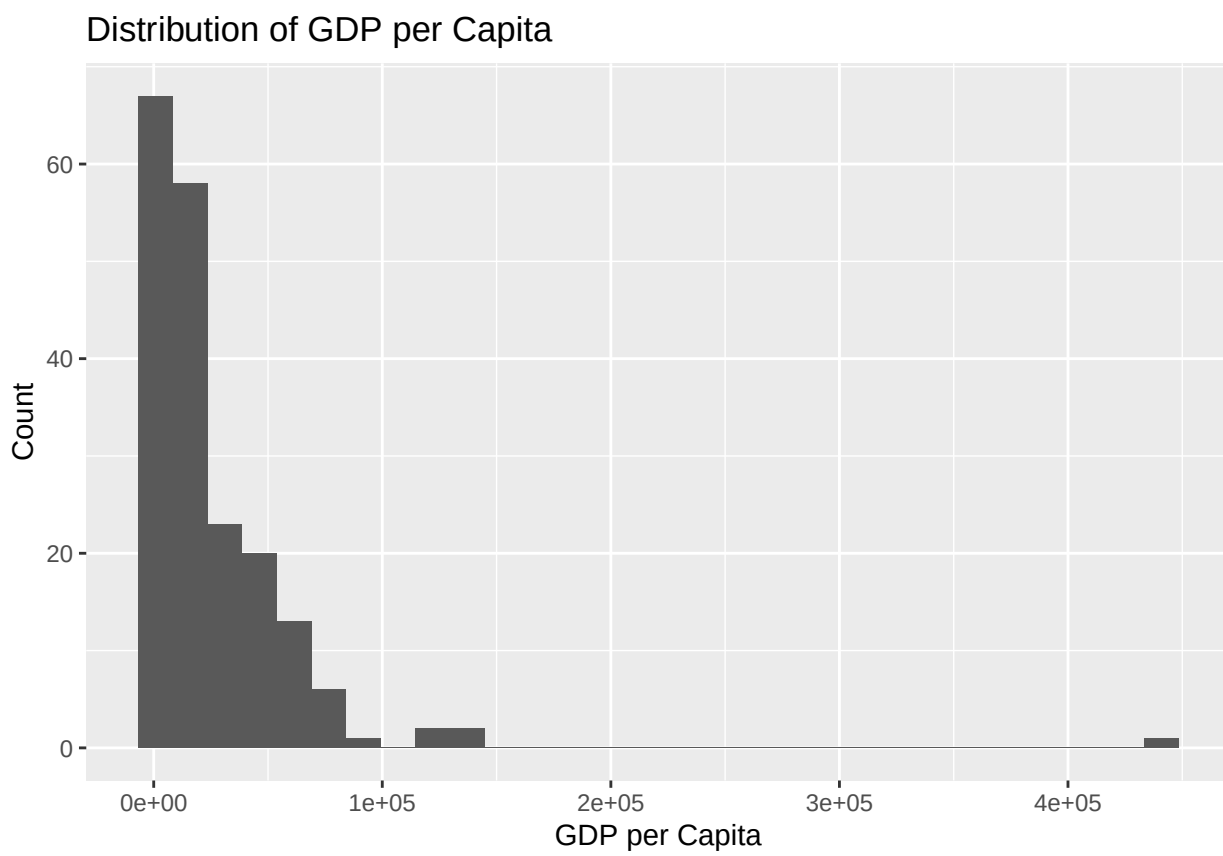
```r
# Use robust standard errors.
robust_se_base <- sqrt(diag(vcovHC(model_base, type = "HC1")))

coeftest(model_base, vcov = vcovHC(model_base, type = "HC1"))
```

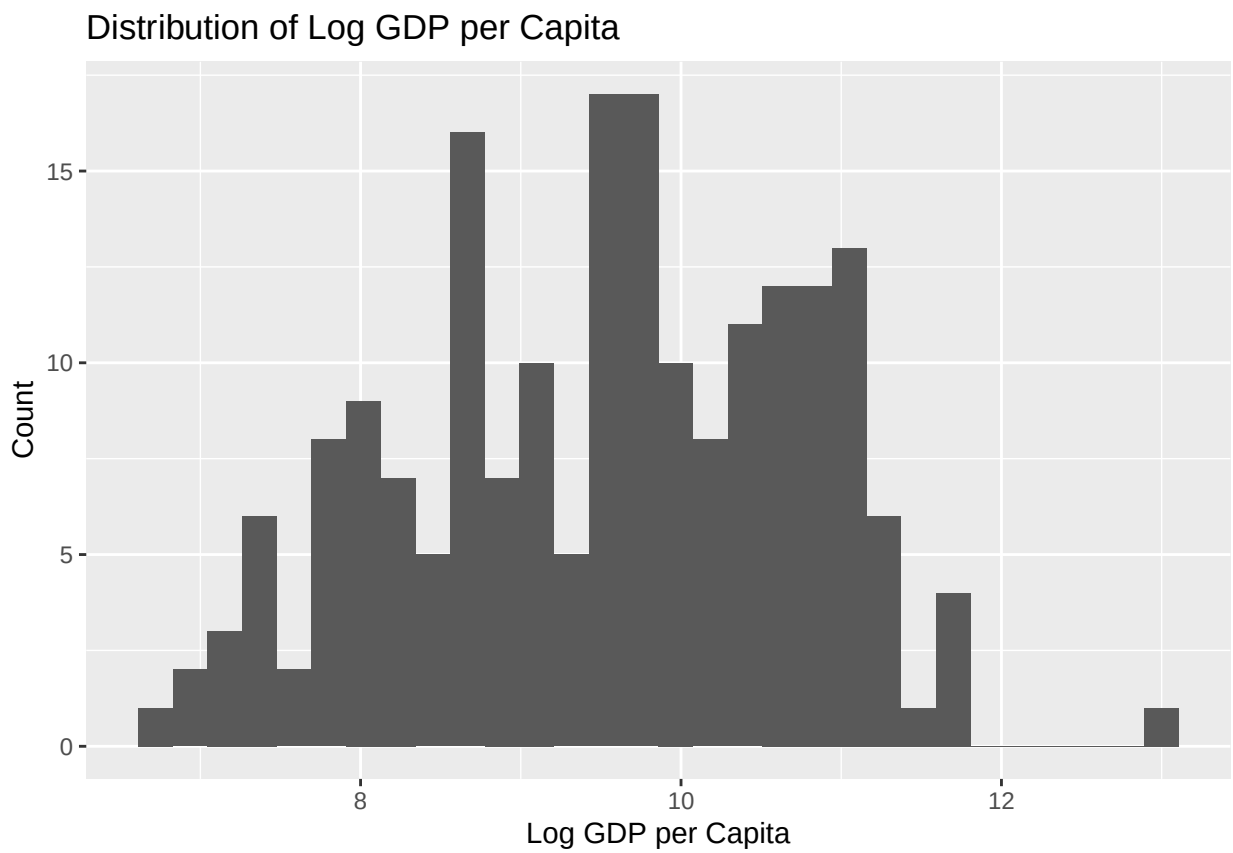
```
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) 6.9657e+01 1.1024e+00 63.1889  < 2e-16 ***
## gdp_pcap    9.5385e-05 4.4195e-05  2.1583  0.03215 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Check for skewness in distribution of gdp per capita.

```
ggplot(df_clean, aes(x = gdp_pcap)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of GDP per Capita",
    x = "GDP per Capita",
    y = "Count"
  )
```

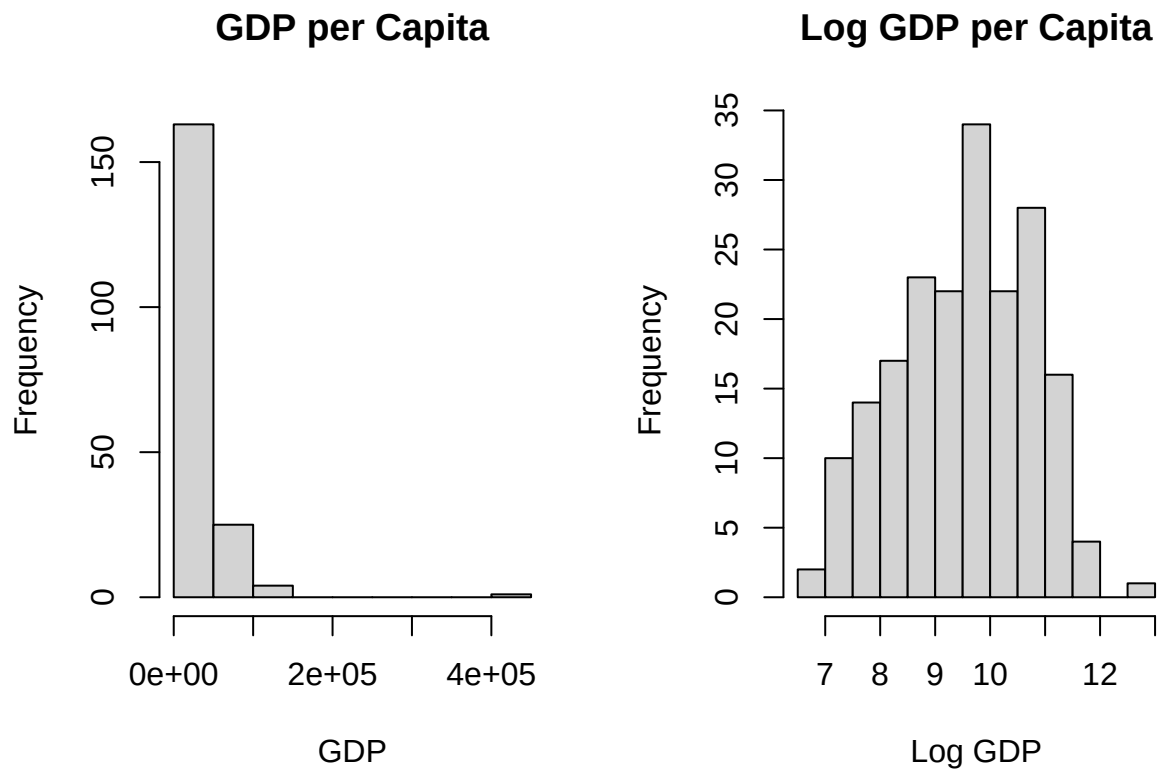



```
ggplot(df_clean, aes(x = log(gdp_pcap))) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of Log GDP per Capita",
    x = "Log GDP per Capita",
    y = "Count"
  )
```




```r
par(mfrow = c(1, 2))

hist(df_clean$gdp_pcap, main = "GDP per Capita", xlab = "GDP")
hist(log(df_clean$gdp_pcap), main = "Log GDP per Capita", xlab = "Log GDP")
```



High correlation between gdp per capita and average daily income. Exclude the latter variable.

```
cor(df_clean$gdp_pcap, df_clean$average_daily_income, use = "complete.obs")
```

```
## [1] 0.9602411
```

```
cor(log(df_clean$gdp_pcap), df_clean$average_daily_income, use = "complete.obs")
```

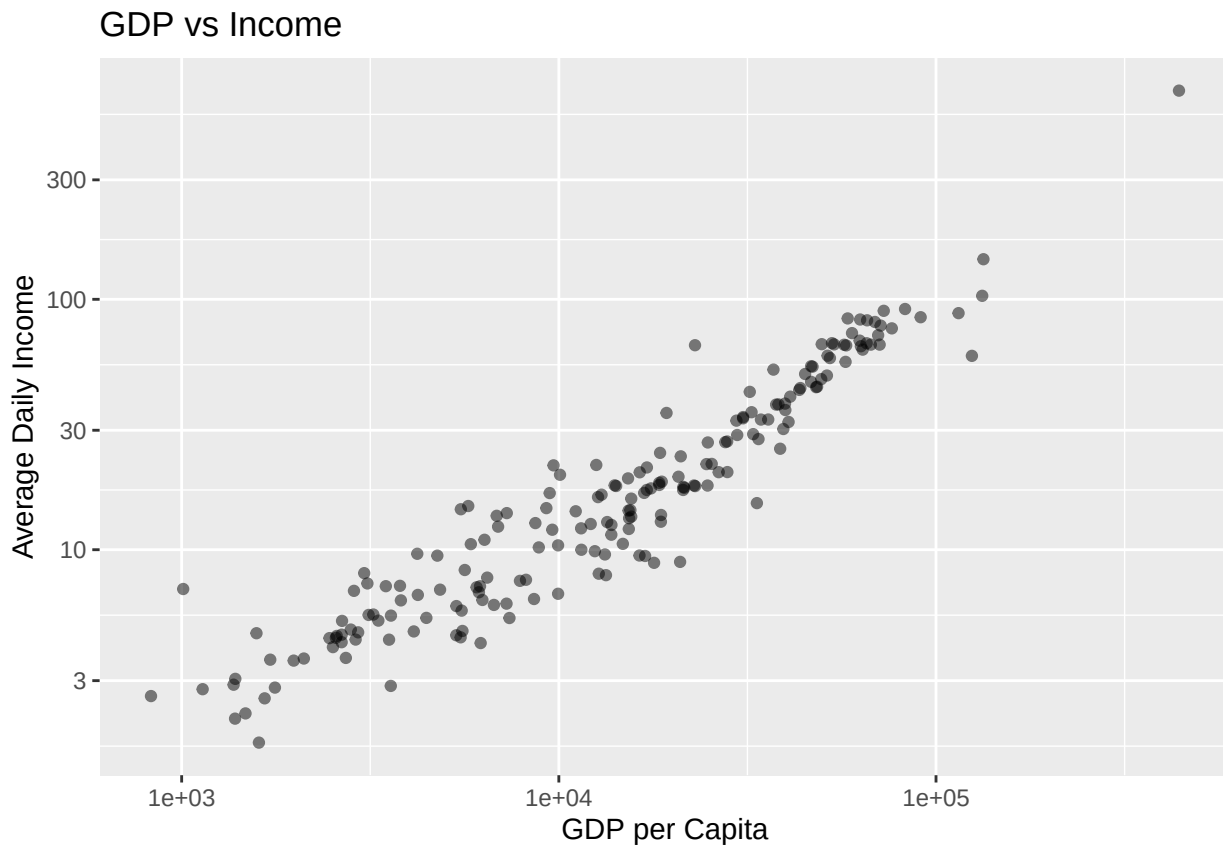
```
## [1] 0.5658659
```

```
cor(log(df_clean$gdp_pcap), log(df_clean$average_daily_income), use = "complete.obs")
```

```
## [1] 0.9477717
```

```
# Plot gdp per capita vs average daily income.
ggplot(df_clean, aes(x = gdp_pcap, y = average_daily_income)) +
  geom_point(alpha = 0.5) +
  scale_x_log10() +
  scale_y_log10() +
  labs(
    x = "GDP per Capita",
```

```
    y = "Average Daily Income",
    title = "GDP vs Income"
  )
```



Correlation between life expectancy and log GDP is 0.796.

```
cor(log(df_clean$gdp_pcap), df_clean$lex, use = "complete.obs")
```

```
## [1] 0.7957497
```

```
model_log <- lm(lex ~ log(gdp_pcap), data = df_clean)
summary(model_log)
```

```
##
## Call:
## lm(formula = lex ~ log(gdp_pcap), data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.4252  -2.8768   0.7347   3.4540  12.3755
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    26.5006     2.5353   10.45   <2e-16 ***
## log(gdp_pcap)   4.7963     0.2641   18.16   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.414 on 191 degrees of freedom
## Multiple R-squared:  0.6332, Adjusted R-squared:  0.6313
## F-statistic: 329.7 on 1 and 191 DF,  p-value: < 2.2e-16
```

```r
# Use robust standard errors.
robust_se_log <- sqrt(diag(vcovHC(model_log, type = "HC1")))

coeftest(model_log, vcov = vcovHC(model_log, type = "HC1"))
```

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)   26.50065    2.70927  9.7815 < 2.2e-16 ***
## log(gdp_pcap)  4.79633    0.27998 17.1311 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Add demographic additional variables.

```r
model_3 <- lm(
  lex ~ log(gdp_pcap) +
    children_per_woman_total_fertil +
    median_age_years,
    data = df_clean
)

summary(model_3)
```

```
##
## Call:
## lm(formula = lex ~ log(gdp_pcap) + children_per_woman_total_fertil +
##     median_age_years, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.5012  -2.4473   0.4313   2.8813   8.2074
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    46.32660    4.60465  10.061  < 2e-16 ***
## log(gdp_pcap)                   2.41537    0.46081   5.242 4.23e-07 ***
## children_per_woman_total_fertil -1.27426    0.46746  -2.726  0.00702 **
## median_age_years                0.20498    0.06958   2.946  0.00362 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.052 on 189 degrees of freedom
## Multiple R-squared:  0.6942, Adjusted R-squared:  0.6894
## F-statistic:   143 on 3 and 189 DF,  p-value: < 2.2e-16
```

```r
anova(model_log, model_3)
```

```
## Analysis of Variance Table
##
## Model 1: lex ~ log(gdp_pcap)
## Model 2: lex ~ log(gdp_pcap) + children_per_woman_total_fertil + median_age_years
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    191 3721.5
## 2    189 3102.7  2    618.83 18.848 3.438e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Use robust standard errors.
coeftest(model_3, vcov = vcovHC(model_3, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                  Estimate Std. Error t value  Pr(>|t|)
## (Intercept)                     46.326603   5.259005  8.8090 8.088e-16 ***
## log(gdp_pcap)                    2.415374   0.531612  4.5435 9.849e-06 ***
## children_per_woman_total_fertil -1.274257   0.479908 -2.6552  0.008602 **
## median_age_years                 0.204985   0.077897  2.6315  0.009203 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Add inequality additional variables.

```r
model_4 <- lm(
  lex ~ log(gdp_pcap) +
    children_per_woman_total_fertil +
    median_age_years +
    gini +
    population_density_per_square_k,
    data = df_clean
)

summary(model_4)
```

```
##
## Call:
## lm(formula = lex ~ log(gdp_pcap) + children_per_woman_total_fertil +
##     median_age_years + gini + population_density_per_square_k,
##     data = df_clean)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.0370  -2.5192   0.5927   2.8748   8.5220
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     56.0855161  5.4926789  10.211  < 2e-16 ***
## log(gdp_pcap)                    2.2994919  0.4585445   5.015 1.23e-06 ***
## children_per_woman_total_fertil -1.4801855  0.4753091  -3.114 0.002135 **
## median_age_years                 0.1411200  0.0706085   1.999 0.047097 *
## gini                            -0.1570525  0.0409944  -3.831 0.000174 ***
## population_density_per_square_k -0.0000723  0.0001541  -0.469 0.639390
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.916 on 187 degrees of freedom
## Multiple R-squared:  0.7173, Adjusted R-squared:  0.7098
## F-statistic: 94.91 on 5 and 187 DF,  p-value: < 2.2e-16
```

```r
anova(model_3, model_4)
```

```
## Analysis of Variance Table
##
## Model 1: lex ~ log(gdp_pcap) + children_per_woman_total_fertil + median_age_years
## Model 2: lex ~ log(gdp_pcap) + children_per_woman_total_fertil + median_age_years +
##     gini + population_density_per_square_k
##   Res.Df    RSS Df Sum of Sq     F    Pr(>F)
## 1    189 3102.7
## 2    187 2868.0  2    234.72 7.652 0.0006394 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Use robust standard errors.
coeftest(model_4, vcov = vcovHC(model_4, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                    Estimate  Std. Error t value  Pr(>|t|)
## (Intercept)                      5.6086e+01  6.3368e+00  8.8508 6.564e-16 ***
## log(gdp_pcap)                    2.2995e+00  5.6341e-01  4.0814 6.625e-05 ***
## children_per_woman_total_fertil -1.4802e+00  4.6506e-01 -3.1828  0.001709 **
## median_age_years                 1.4112e-01  7.5910e-02  1.8591  0.064591 .
## gini                            -1.5705e-01  4.7804e-02 -3.2853  0.001216 **
## population_density_per_square_k -7.2303e-05  1.3410e-04 -0.5392  0.590414
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
model_5 <- lm(
  lex ~ log(gdp_pcap) +
    children_per_woman_total_fertil +
```

```r
    median_age_years +
    gini +
    log(population_density_per_square_k),
    data = df_clean
)

summary(model_5)
```

```
##
## Call:
## lm(formula = lex ~ log(gdp_pcap) + children_per_woman_total_fertil +
##     median_age_years + gini + log(population_density_per_square_k),
##     data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.0807  -2.6426   0.7079   2.8211   7.8085
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         53.77851    5.42558   9.912  < 2e-16 ***
## log(gdp_pcap)                        2.33868    0.44575   5.247 4.17e-07 ***
## children_per_woman_total_fertil     -1.44067    0.45610  -3.159 0.001849 **
## median_age_years                     0.13039    0.06903   1.889 0.060442 .
## gini                                -0.14709    0.04095  -3.592 0.000419 ***
## log(population_density_per_square_k)  0.39095    0.20653   1.893 0.059913 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.882 on 187 degrees of freedom
## Multiple R-squared:  0.7223, Adjusted R-squared:  0.7149
## F-statistic: 97.29 on 5 and 187 DF,  p-value: < 2.2e-16
```

```r
# Use robust standard errors.
coeftest(model_5, vcov = vcovHC(model_5, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                      Estimate Std. Error t value  Pr(>|t|)
## (Intercept)                         53.778508   6.920919  7.7704 5.041e-13 ***
## log(gdp_pcap)                        2.338677   0.545385  4.2881 2.884e-05 ***
## children_per_woman_total_fertil     -1.440669   0.500542 -2.8782  0.004465 **
## median_age_years                     0.130390   0.075074  1.7368  0.084063 .
## gini                                -0.147090   0.048589 -3.0272  0.002816 **
## log(population_density_per_square_k)  0.390949   0.231887  1.6859  0.093474 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Add Tech/Infrastructure additional variables.

```r
model_6 <- lm(
  lex ~ log(gdp_pcap) +
    children_per_woman_total_fertil +
    median_age_years +
    gini +
    log(population_density_per_square_k) +
    cell_phones_per_100_people +
    broadband_subscribers_per_100_p +
    co2_pcap_cons,
    data = df_clean
)

summary(model_6)
```

```
##
## Call:
## lm(formula = lex ~ log(gdp_pcap) + children_per_woman_total_fertil +
##     median_age_years + gini + log(population_density_per_square_k) +
##     cell_phones_per_100_people + broadband_subscribers_per_100_p +
##     co2_pcap_cons, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.3738  -2.6355   0.4774   2.8431   6.9417
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         49.229105   7.827099   6.290 2.42e-09 ***
## log(gdp_pcap)                        2.872687   0.698406   4.113 5.97e-05 ***
## children_per_woman_total_fertil     -1.227179   0.551897  -2.224  0.02744 *
## median_age_years                     0.041032   0.090197   0.455  0.64973
## gini                                -0.135688   0.041950  -3.235  0.00145 **
## log(population_density_per_square_k)  0.552802   0.222442   2.485  0.01388 *
## cell_phones_per_100_people          -0.002326   0.010424  -0.223  0.82366
## broadband_subscribers_per_100_p      0.070823   0.042255   1.676  0.09549 .
## co2_pcap_cons                       -0.100178   0.083217  -1.204  0.23027
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.861 on 177 degrees of freedom
##   (7 observations deleted due to missingness)
## Multiple R-squared:  0.7253, Adjusted R-squared:  0.7129
## F-statistic: 58.42 on 8 and 177 DF,  p-value: < 2.2e-16
```

```r
modelsummary(
  list(
    "Baseline" = model_base,
    "Log GDP" = model_log,
    "Demographics" = model_3,
    "Final Model" = model_5
```

Table 1: Regression Models of Life Expectancy

	Baseline	Log GDP	Demographics	Final Model
Intercept	69.657***	26.501***	46.327***	53.779***
	(1.102)	(2.709)	(5.259)	(6.921)
GDP per Capita	0.000*			
	(0.000)			
Log GDP per Capita		4.796***	2.415***	2.339***
		(0.280)	(0.532)	(0.545)
Children per Woman			−1.274**	−1.441**
			(0.480)	(0.501)
Log Population Density				0.391+
				(0.232)
Median Age			0.205**	0.130+
			(0.078)	(0.075)
Gini Index				−0.147**
				(0.049)
Num.Obs.	193	193	193	193
R2	0.267	0.633	0.694	0.722
R2 Adj.	0.264	0.631	0.689	0.715
Std.Errors	HC1	HC1	HC1	HC1

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

```
  ),
  vcov = "HC1",
  stars = TRUE,
  title = "Regression Models of Life Expectancy",
  statistic = "({std.error})",
  gof_omit = "AIC|BIC|Log.Lik|RMSE|F",
  coef_map = c(
    "(Intercept)" = "Intercept",
    "gdp_pcap" = "GDP per Capita",
    "log(gdp_pcap)" = "Log GDP per Capita",
    "children_per_woman_total_fertil" = "Children per Woman",
    "log(population_density_per_square_k)" = "Log Population Density",
    "median_age_years" = "Median Age",
    "gini" = "Gini Index"
  )
)
```